#13 · 3.4% of marks

Ch 4 — Security Analysis

Theory-heavy. SM TYK theory Qs: **EMH and its misconception · Dow theory · Elliot wave theory · indicators of economic performance.** Numericals: moving averages / EMA, RSI (ICAI paper), dividend-growth value.

① Fundamental analysis (SM 1)

- → *Share price depends on **expected future dividends**; PV of dividends = **intrinsic value**. Price below intrinsic → **buy**; above → **sell***
- → *Constant dividend: $P_0 = D \div k$. Constant growth ($g < k$): **$P_0 = D_1 \div (k - g) = D_0(1 + g) \div (k - g)$** — **Gordon's dividend growth model** (Myron J. Gordon); invalid if $g \geq k$ (infinite price) — not restrictive, long-term g unlikely to exceed k. $g = 0$ → back to $D \div k$*
- → ***Link to P/E (SM 1.1)**: with payout fraction **b**, $D_1 = bE_1$ → **$P_0 = bE_1 \div (k - g)$**; so $P/E = b \div (k - g)$. Price / P/E above fundamental → overpriced → sell; below → buy*
- → *Three concentric circles: **economy** → **industry** → **company** analysis*

> ⚠ **TRAP**
> ***Symbol trap**: in SM Ch 4, **b = payout** ratio. In SM Ch 5 (Gordon earnings model) and Ch 1 (SGR), **b = retention**. Read the definition given in the question.*

② Economic analysis (SM 1.2)

Past / present / expected performance of the economy and sectors; income & consumption trends drive profits, dividends, prices.

Factors (4)

*(a) growth of **national income** — nominal GDP vs **real** growth (b) growth of **industrial sector** (by demand) (c) **inflation** — WPI / CPI; consumer industries fine-tune pricing (d) **monsoon** — forward & backward linkages*

Techniques (3) — TYK Q4

*(a) **Anticipatory surveys** — expert opinion on construction, capex, inventory, consumer spending; drawbacks: intentions may not materialise; not forecasts per se (b) **Barometer / indicator approach** (c) **Economic model building** (GNP model / sectoral analysis)*

- → ***Indicators**: **leading** (turn before the economy) · **roughly coincidental** (turn with it) · **lagging** (turn after it). Show direction, not magnitude; may conflict → **diffusion / composite index** (hard to compute; doesn't remove irregular movements). **Money supply** ↑ → inflation; stocks a hedge → prices rise*
- → ***GNP model steps**: (i) hypothesise total demand (GNP) from political stability, inflation, economic levels (ii) forecast components — consumption, gross private domestic investment, government purchases, net exports (iii) add them (iv) compare with an independent agency's forecast and test consistency*

THEORY

③ Industry analysis (SM 1.3)

Industries don't grow at the same rate → assess demand, cost structure, government constraints.

Factor (7)	SM
(a) Product life cycle	High profit in initial & growth stages, medium but steady in maturity, sharp decline at the end
(b) Demand-supply gap	Excess supply cuts profit (lower price); shortage raises it
(c) Barriers to entry	Innate to product / technology, or created by existing firms
(d) Government attitude	Crucial determinant of prospects
(e) State of competition	Leaders, homogeneous vs differentiated products, demand by customer / area, growth / cyclical / defensive / decline industry
(f) Cost conditions & profitability	Cost allocation & controllability, price, capacity (installed / idle / operating), capex needed; ratios — GP, OPM, ROE, return on capital
(g) Technology & research	Nature of technology, expected changes, capex-to-sales over time, R&D spend, sales & profit in short / immediate / long run

Techniques: (a) **regression & correlation** on demand drivers (GNP, disposable income, per-capita consumption, price elasticity) (b) **input-output analysis** — flow of goods through the economy to spot growing / declining industries.

④ Company analysis (SM 1.4) — 13 points

- **(a) Net worth & book value** — equity + preference + free reserves − intangibles − losses; BV may not reflect worth: price reflects **future earnings**; BV on **historical cost**
- **(b) Sources & uses (funds flow)** — short-term funds financing long-term assets → liquidity crunch
- **(c) Cross-sectional & time series** — common-size statements, ratio analysis
- **(d) Size & ranking** — capital employed, profit, ROI, sales vs peers; know-how, R&D, price leadership
- **(e) Growth record — P/E**, % growth in EPS, % growth in **net block** (capacity). SM: A ₹150 / EPS 25 = P/E 6 vs B ₹250 / 25 = 10 → A preferable if all else equal. "**Growth is the single most important factor**"
- **(f) Financial analysis** — key: EPS, P/E, yield, book value, intrinsic value (10 years, vs industry); solvency (current, liquid, D/E)
- **(g) Competitive advantage** (Apple's brand, Samsung in mobiles) · **(h) Quality of management** — experience, skills, integrity; earnings calls, MD&A, insider selling, 5-year record · **(i) Corporate governance** — LODR 2015 compliance, stakeholder policies, disclosure quality & timeliness, independent directors · **(j) Regulation** — Companies Act, SCRA, SEBI Act; FEMA / RBI · **(k) Location & labour relations** · **(l) Pattern of shareholding** (PNB nationalisation — institutions raised compensation) · **(m) Marketability** — public holding, speculative interest, exchange, volume

Techniques: correlation & regression (simple / multiple) · trend analysis · decision tree (multiply probabilities along each sequence, then add). Also **group analysis** — SEBI disclosure of group companies; RBI group exposure norms.

TYK THEORY Q2 & Q3

5 Technical analysis (SM 2.1–2.3)

- → Studies price charts assuming trends **repeat** (investor psychology follows patterns); studies investor and price behaviour. Asks: (i) is there a trend? (ii) will it reverse? Methods: **visual** (charts) and **statistical**
- → **Assumptions (4)**: (i) value depends on **supply & demand** (ii) S & D governed by rational and irrational factors (iii) prices move in **trends** lasting a substantial time (iv) relies on charts, not financial statements
- → **Principles (3)**: (a) **market discounts everything** (b) **price moves in trends** (c) **history repeats itself**

6 Dow theory (SM 2.4.1) — TYK Q2

Charles Dow (founder of Dow Jones), late 19th century; measures relative strength of the market; a business barometer. Based on **DJIA** and **DJTA**. Three simultaneous movements: **primary** (main bull / bear trend, 1 year to 36 months or longer) · **secondary** (opposite direction, 2 weeks to a month or more) · **daily** fluctuations (not interpreted, but studied). Tells **where the market is going, not how far**. Successively higher highs & higher lows → bull; lower highs & lows → bear.

Three moves up: (1) accumulation by far-sighted investors (2) first reports of good earnings (3) widespread reports of well-being + rampant speculation. Then far-sighted investors sell → first move down; non-sustainability confirmed → second; **distress selling** → third.

7 Elliot wave theory (SM 2.4.2) — TYK Q3

Ralph Elliot, 1934, inspired by Dow theory and nature; 75 years of price data. Wave = price movement from one change in direction to the next in the same direction, from buying / selling impulses. **Impulsive (basic) waves**: 3 or 5 waves in the main direction (bull or bear). **Corrective (reaction) waves**: 3 waves against it. Waves 1, 3, 5 directional; 2 & 4 corrective. **Complete cycle = 5-wave impulse + 3-wave correction (a, b, c) = 8 waves**, bullish and bearish phases; then a fresh cycle.

8 Random walk theory (SM 2.4.3)

Prices are statistical ups and downs, **unpredictable**; no link between present and future prices; successive changes independent. **M. G. Kendall** (SM: "Kendell") — like a roulette wheel; "drunk walking in a blind lane". (a) prices can't be predicted (b) trends are a statistical expression of past data (c) no connection between successive peaks and troughs.

9 Charts (SM 2.5)

Line (closing prices joined) · **Bar** (vertical high-low bar; ticks for open & close) · **Japanese candlestick** (white = close > open = bullish; black = close < open = bearish; **Doji** = open ≈ close) · **Point & figure** (detects reversals; choose **box size** — smaller = more sensitive — and **reversal criterion**; X rising, O falling).

THEORY + NUMERICAL

10 Market indicators (SM 2.6)

- → **Breadth index** = net advances or declines ÷ issues traded; supports Dow averages = technical strength; contradicts = weakness (reversal)
- → **Volume**: rising price + rising volume → buy (unsatisfied demand); falling + rising volume → bear. SM also: rising market with decreasing volume indicates a bull market; falling with dwindling volume a bear market. Use with Dow theory
- → **Confidence index** = high-grade bond yields ÷ low-grade bond yields; rising precedes a rising market; usually (not always) a **leading** indicator
- → **Relative strength**: some securities rise faster in bull / fall slower in bear and keep that strength — measured by historical returns or ratio to industry / market
- → **Odd-lot theory**: **contrary-opinion** — average person usually wrong; predicts tops in bull markets and reversals

11 Support, resistance & patterns (SM 2.7–2.8)

Peak after a fall = **resistance** (selling pressure); trough after a rebound = **support** (buying pressure); breach = new levels. SM eg: A ₹100 / ₹125; B ₹140 / ₹160.

Pattern	Signal
Channel	Uniform tops & bottoms; up-sloping = rising, down-sloping = falling
Wedge	Tops and bottoms move in opposite directions, or same direction at different rates
Head & shoulders (top)	Most important **reversal** pattern; **bearish** — below neckline → fall ≈ head-to-neckline distance → **sell**
Inverse head & shoulders	**Bullish** — above neckline → **buy**
Triangle / coil	Uncertainty; breakout direction hard to predict
Flags & pennants	Previous trend likely to **continue**
Double top / double bottom	Bearish / bullish
Gap	Open vs previous close; wider gap = stronger continuation signal

12 Moving averages (SM 2.9)

$AMA_{n,t}$ = $(P_t + \dots + P_{t-(n-1)}) \div n$. Trend: **200-day** (or 30-week) = long term · **60-day** = intermediate · **10-day** = short term

$EMA_t = a \times P_t + (1 - a) \times EMA_{t-1}$, $a = 2 \div (n + 1)$ — latest price weighted most

Buy signal	Sell signal
(a) price rises through a flattening MA (b) price falls below a **rising** MA (c) price above MA falls but turns up before reaching it	(a) price falls through a flattening MA (b) price rises above a **falling** MA (c) price below MA rises but turns down before reaching it

ICAI paper (RSI): RSI = 100 − 100 ÷ (1 + RS), RS = avg gain ÷ avg loss; update averages as (previous × 13 + today) ÷ 14. >70 overbought · 50–70 bullish · 30–50 bearish · <30 oversold (ICAI answer: 34.64 → bearish).

TYK THEORY Q1

13 Evaluation of technical analysis (SM 2.10) & FA vs TA (SM 3)

For (3)

(a) crowd psychology makes trends persist; spotted early (b) demand-supply shifts are gradual (c) information is assimilated over time, so price keeps moving one way

Against (4)

(a) no convincing explanation of tools (b) evidence for random walk (c) by the time a trend is signalled it has happened (d) **self-defeating** — more users, less value

Efficient market → TA may not work; real markets are imperfect → useful **with** fundamental analysis. **FA vs TA**: method — macro (GDP, inflation, rates) + micro (sales, profitability, solvency) vs historical market data (price, volume, open interest) · rule — "prices discount everything" vs "price captures everything" · use — **long-term** vs **short-term** investing.

14 Efficient market theory (SM 4) — TYK Q1

- **Eugene Fama** (Chicago), 1960s: all price-sensitive information is fully reflected in prices → no one can consistently beat the market; higher return only by taking **more risk**
- **Why random**: information free & instant to all · keen competition makes prices reflect intrinsic values · prices change only on **new, unpredictable** information
- **Misconceptions (SM 4.2)**: (1) EMH doesn't mean prices always equal fair value — the future is uncertain, surprises keep prices fluctuating (2) managers' failure to outperform doesn't mean incompetence — markets are efficient **because** managers compete well (3) random movement ≠ irrational market — **rational, competitive investors make price changes random**
- **Levels**: **weak** — past prices & volumes · **semi-strong** — + all public information · **strong** — all information, public and private
- **Weak-form tests**: **serial correlation** (none found, short-term) · **run test** (runs vs a random series — supports random walk) · **filter rule** (buy after N% rise, sell after N% fall — doesn't beat buy-and-hold after commissions)
- **Semi-strong**: Fama, Fisher, Jensen & Roll — 940 NYSE splits 1957–59; Ball & Brown (SM: "Boll") — earnings announcements. **Anomalies**: slow adjustment to quarterly earnings surprises · small firms beat large · low P/E beats high P/E · **Monday** returns lower
- **Strong**: insiders and exchange specialists earn superior returns; **mutual fund managers do not** (Burton Malkiel, *A Random Walk Down Wall Street*)
- **Challenges (4)**: (a) information inadequacy / misinformation (b) limited processing capacity (Herbert Simon; David Dreman) (c) irrational behaviour (Keynes; **L. C. Gupta** — "blind man firing a gun") (d) monopolistic influence of big institutions

Equity research (SM 5): buy / sell / hold calls, also used in M&A swap ratios. Tools: Bloomberg Terminal (1981), Benzinga Pro, Refinitiv Eikon, MarketXLS, Stockopedia, Koyfin, Finbox.io, GuruFocus, Business Quant, YCharts.

15 Correlation — how Ch 4 actually connects

Ch 4 idea	The actual link (app-written cross-reference; the Ch 4 side is SM)
Intrinsic value, Gordon	**Ch 5**: same DDM; P/E = payout ÷ (k − g). **Ch 14**: equity research feeds the swap ratio.
EMH levels	**Ch 6 / Ch 8**: efficient markets favour **passive** (index / ETF) strategies; MF managers not beating the market = strong-form evidence; Jensen's α measures any "beating".
Leading indicators, money supply, inflation	**Ch 2**: interest-rate & currency risk identifiers; **Ch 3**: government-policy impact on capital budgeting.
Odd-lot, confidence index	**Ch 5**: confidence index uses bond yields — high-grade vs low-grade = credit spread.

④ Security Analysis — what ICAI asked (Sep 2025, Jan 2026, May 2026)

12 marks *set across the last three attempts (3.4% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	*Q4(c) [either]*	*4*	*Technical analysis: favourable and unfavourable arguments (Theory)*
Jan 2026	*Q4(c)*	*4*	*Technical analysis: identify chart patterns (uptrend, continuation/flag, head and shoulders, triangle/wedge) from descriptions, any four (Theory)*
May 2026	*Q4(c) [either]*	*4*	*Elliott Wave Theory of technical analysis (Theory)*

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Explain the Efficient Market Theory in and what is major misconception about this theory?

ICAI answer: "Please refer paragraph 4" – SM paragraph 4 reproduced

4. EFFICIENT MARKET THEORY (EFFICIENT MARKET HYPOTHESIS)

Efficient Market Theory was developed by University of Chicago professor Eugen Fama in the 1960s. As per this theory, at any given time, all available price sensitive information is fully reflected in securities' prices. Thus, this theory implies that no investor can consistently outperform the market as every stock is appropriately priced based on available information.

Stating otherwise this theory states that no one can "beat the market" hence making it impossible for investors to either purchase undervalued stocks or sell stocks for inflated prices as stocks are always traded at their fair value on stock exchanges. Hence it is impossible to outperform the overall market through expert stock selection or market timing and that the only way an investor can possibly obtain higher returns is by purchasing risky investments.

4.1 Search for Theory

When empirical evidence in favour of Random walk hypothesis seemed overwhelming, researchers wanted to know about the Economic processes that produced a Random walk. They concluded that randomness of stock price was a result of efficient market that led to the following view points:

- Information is freely and instantaneously available to all market participants.
- Keen competition among the market participants more or less ensures that market will reflect intrinsic values. This means that they will fully impound all available information.
- Price change only response to new information that is unrelated to previous information and therefore unpredictable.

4.2 Misconception about Efficient Market Theory

Efficient Market Theory implies that market prices factor in all available information and as such it is not possible for any investor to earn consistent long term returns from market operations.

Although price tends to fluctuate they cannot reflect fair value. This is because the future is uncertain. The market springs surprises continually and as prices reflect the surprises they fluctuate.

Inability of institutional portfolio managers to achieve superior investment performance implies that they lack competence in an efficient market. It is not possible to achieve superior investment performance since market efficiency exists due to portfolio mangers doing this job well in a competitive setting.

The random movement of stock prices suggests that stock market is irrational. Randomness and irrationality are two different things, if investors are rational and competitive, price changes are bound to be random.

4.3 Level of Market Efficiency

That price reflects all available information, the highest order of market efficiency. According to Eugene Fama, there exist three levels of market efficiency:-

(i) *Weak form efficiency* – Price reflects all information found in the record of past prices and volumes.

(ii) *Semi – Strong efficiency* – Price reflects not only all information found in the record of past prices and volumes but also all other publicly available information.

(iii) *Strong form efficiency* – Price reflects all available information public as well as private.

4.4 Empirical Evidence on Weak form of Efficient Market Theory

According to the Weak form Efficient Market Theory current price of a stock reflect all information found in the record of past prices and volumes. This means that there is no relationship between the past and future price movements.

Three types of tests can be employed to empirically verify the weak form of Efficient Market Theory- Serial Correlation Test, Run Test and Filter Rule Test.

(a) *Serial Correlation Test*: To test for randomness in stock price changes, one has to look at serial correlation. For this purpose, price change in one period has to be correlated with price change in some other period. Price changes are considered to be serially independent. Serial correlation studies employing different stocks, different time lags and different time period have been conducted to detect serial correlation but no significant serial correlation could be discovered. These studies were carried on short term trends viz. daily, weekly, fortnightly and monthly and not in long term trends in stock prices as in such cases. Stock prices tend to move upwards.

(b) *Run Test*: Given a series of stock price changes each price change is designated + if it represents an increase and – if it represents a decrease. The resulting series may be -,+, - , -, - , +, +.

A run occurs when there is no difference between the sign of two changes. When the sign of change differs, the run ends and new run begins.

$$\underset{1}{++}\,/\,\underset{2}{---}\,/\,\underset{3}{+}\,/\,\underset{4}{-}\,/\,\underset{5}{+}\,/\,\underset{6}{--}\;\underset{1}{++}\,/\,\underset{2}{---}\,/\,\underset{3}{+}\,/\,\underset{4}{-}\,/\,\underset{5}{+}\,/\,\underset{6}{--}$$

To test a series of price change for independence, the number of runs in that series is compared with a number of runs in a purely random series of the size and in the process determines whether it is statistically different. By and large, the result of these studies strongly supports the Random Walk Model.

(c) *Filter Rules Test*: If the price of stock increases by at least N% buy and hold it until its price decreases by at least N% from a subsequent high. When the price decreases at least N% or more, sell it. If the behaviour of stock price changes is random, filter rules should not apply in such a buy and hold strategy. By and large, studies suggest that filter rules do not out perform a single buy and hold strategy particular after considering commission on transaction.

4.5 Empirical Evidence on Semi-strong Efficient Market Theory

Semi-strong form efficient market theory holds that stock prices adjust rapidly to all publicly available information. By using publicly available information, investors will not be able to earn above normal rates of return after considering the risk factor. To test semi-strong form efficient market theory, a

number of studies was conducted which lead to the following queries: Whether it was possible to earn on the above normal rate of return after adjustment for risk, using only publicly available information and how rapidly prices adjust to public announcement with regard to earnings, dividends, mergers, acquisitions, stock-splits?

Several studies support the Semi-strong form Efficient Market Theory. Fama, Fisher, Jensen and Roll in their adjustment of stock prices to new information examined the effect of stock split on return of 940 stock splits in New York Stock Exchange during the period 1957-1959 They found that prior to the split, stock earns higher returns than predicted by any market model.

Boll and Brown in an empirical evaluation of accounting income numbers studied the effect of annual earnings announcements. They divided the firms into two groups. First group consists of firms whose earnings increased in relation to the average corporate earnings while second group consists of firms whose earnings decreased in relation to the average corporate earnings. They found that before the announcement of earnings, stock in the first group earned positive abnormal returns while stock in the second group earned negative abnormal returns after the announcement of earnings. Stock in both the groups earned normal returns.

There have been studies which have been empirically documented showing the following inefficiencies and anomalies:

- Stock price adjust gradually not rapidly to announcements of unanticipated changes in quarterly earnings.
- Small firms' portfolio seemed to outperform large firms' portfolio.
- Low price earning multiple stock tend to outperform high price earning multiple stock.
- Monday's return is lower than return for the other days of the week.

4.6 Empirical Evidence on Strong form of Efficient Market Theory

According to the Efficient Market Theory, all available information, public or private, is reflected in the stock prices. This represents an extreme hypothesis.

To test this theory, the researcher analysed returns earned by certain groups viz. corporate insiders, specialists on stock exchanges, mutual fund managers who have access to internal information (not publicly available), or possess greater resource or ability to intensively analyse information in the public domain. They suggested that corporate insiders (having access to internal information) and stock exchange specialists (having monopolistic exposure) earn superior rate of return after adjustment of risk.

Mutual Fund managers do not on an average earn a superior rate of return. No scientific evidence has been formulated to indicate that investment performance of professionally managed portfolios as a group has been any better than that of randomly selected portfolios. This was the finding of Burton Malkiel in his Random Walk Down Wall Street, New York.

4.7 Challenges to the Efficient Market Theory

(a) Information inadequacy – Information is neither freely available nor rapidly transmitted to all participants in the stock market. There is a calculated attempt by many companies to circulate misinformation.

(b) Limited information processing capabilities – Human information processing capabilities are sharply limited. According to Herbert Simon every human organism lives in an environment which generates millions of new bits of information every second but the bottlenecks of the perceptual apparatus does not admit more than thousand bits per seconds and possibly much less.

David Dreman maintained that under conditions of anxiety and uncertainty, with a vast interacting information grid, the market can become a giant.

(c) Irrational Behaviour – It is generally believed that investors' rationality will ensure a close correspondence between market prices and intrinsic values. But in practice this is not true. J. M. Keynes argued that all sorts of consideration enter into the market valuation which is in no way relevant to the prospective yield. This was confirmed by L. C. Gupta who found that the market evaluation processes work haphazardly almost like a blind man firing a gun. The market seems to function largely on hit or miss tactics rather than on the basis of informed beliefs about the long term prospects of individual enterprises.

(d) Monopolistic Influence – A market is regarded as highly competitive. No single buyer or seller is supposed to have undue influence over prices. In practice, powerful institutions and big operators wield great influence over the market. The monopolistic power enjoyed by them diminishes the competitiveness of the market.

ICAI answer: "Please refer paragraph 4" — SM paragraph 4 reproduced

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Inability of institutional portfolio managers to achieve superior investment performance implies that they lack competence in an efficient market. It is not possible to achieve superior investment performance since market efficiency exists due to portfolio mangers doing this job well in a competitive setting.

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To test this theory, the researcher analysed returns earned by certain groups viz. corporate insiders, specialists on stock exchanges, mutual fund managers who have access to internal information (not publicly available), or possess greater resource or ability to intensively analyse information in the public domain. They suggested that corporate insiders (having access to internal information) and stock exchange specialists (having monopolistic exposure) earn superior rate of return after adjustment of risk.

Mutual Fund managers do not on an average earn a superior rate of return. No scientific evidence has been formulated to indicate that investment performance of professionally managed portfolios as a group has been any better than that of randomly selected portfolios. This was the finding of Burton Malkiel in his Random Walk Down Wall Street, New York.

4.7 Challenges to the Efficient Market Theory

(a) Information inadequacy – Information is neither freely available nor rapidly transmitted to all participants in the stock market. There is a calculated attempt by many companies to circulate misinformation.

(b) Limited information processing capabilities – Human information processing capabilities are sharply limited. According to Herbert Simon every human organism lives in an environment which generates millions of new bits of information every second but the bottlenecks of the perceptual apparatus does not admit more than thousand bits per seconds and possibly much less.

David Dreman maintained that under conditions of anxiety and uncertainty, with a vast interacting information grid, the market can become a giant.

(c) Irrational Behaviour – It is generally believed that investors' rationality will ensure a close correspondence between market prices and intrinsic values. But in practice this is not true. J. M. Keynes argued that all sorts of consideration enter into the market valuation which is in no way relevant to the prospective yield. This was confirmed by L. C. Gupta who found that the market evaluation processes work haphazardly almost like a blind man firing a gun. The market seems to function largely on hit or miss tactics rather than on the basis of informed beliefs about the long term prospects of individual enterprises.

(d) Monopolistic Influence – A market is regarded as highly competitive. No single buyer or seller is supposed to have undue influence over prices. In practice, powerful institutions and big operators wield great influence over the market. The monopolistic power enjoyed by them diminishes the competitiveness of the market.

Theory Q2

2. Explain Dow theory.

ICAI answer: "Please refer paragraph 2.4.1" – SM paragraph 2.4.1 reproduced

2.4.1 The Dow Theory

The Dow Theory is one of the oldest and most famous technical theories. It was originated by Charles Dow, the founder of Dow Jones Company in late nineteenth century. It is a helpful tool for determining the relative strength of the stock market. It can also be used as a barometer of business.

The Dow Theory is based upon the movements of two indices, constructed by Charles Dow, Dow Jones Industrial Average (DJIA) and Dow Jones Transportation Average (DJTA). These averages reflect the aggregate impact of all kinds of information on the market. The movements of the market are divided into three classifications, all going at the same time; the primary movement, the secondary movement, and the daily fluctuations. The primary movement is the main trend of the market, which lasts from one year to 36 months or longer. This trend is commonly called bear or bull market. The secondary movement of the market is shorter in duration than the primary movement, and is opposite in direction. It lasts from two weeks to a month or more. The daily fluctuations are the narrow movements from day-to-day. These fluctuations are not part of the Dow Theory interpretation of the stock market. However, daily movements must be carefully studied, along with primary and secondary movements, as they go to make up the longer movement in the market.

Thus, the Dow Theory's purpose is to determine where the market is and where is it going, although not how far or high. The theory, in practice, states that if the cyclical swings of the stock market averages are successively higher and the successive lows are higher, then the market trend is up and a bullish market exists. Contrarily, if the successive highs and successive lows are lower, then the direction of the market is down and a bearish market exists.

Charles Dow proposed that the primary uptrend would have three moves up, the *first* one being caused by accumulation of shares by the far-sighted, knowledgeable investors, the *second* move would be caused by the arrival of the first reports of good earnings by corporations, and the last move up would be caused by widespread report of financial well-being of corporations. The *third* stage would also see rampant speculation in the market. Towards the end of the third stage, the far-sighted investors, realizing that the high earnings levels may not be sustained, would start selling, starting the first move down of a downtrend, and as the non-sustainability of high earnings is

confirmed, the second move down would be initiated and then the third move down would result from distress selling in the market.

ICAI answer: "Please refer paragraph 2.4.1" – SM paragraph 2.4.1 reproduced

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Theory Q3

3. Explain the Elliot Wave Theory of technical analysis.

ICAI answer: "Please refer paragraph 2.4.2" – SM paragraph 2.4.2 reproduced

2.4.2 Elliot Wave Theory

Inspired by the Dow Theory and by observations found throughout nature, Ralph Elliot formulated Elliot Wave Theory in 1934. This theory was based on analysis of 75 years stock price movements and charts. From his studies, he defined price movements in terms of waves. Accordingly, this theory was named Elliot Wave Theory. Elliot found that the markets exhibited certain repeated patterns or waves. As per this theory wave is a movement of the market price from one change in the direction to the next change in the same direction. These waves are resulted from buying and selling impulses emerging from the demand and supply pressures on the market. Depending on the demand and supply pressures, waves are generated in the prices.

As per this theory, waves can be classified into two parts:-

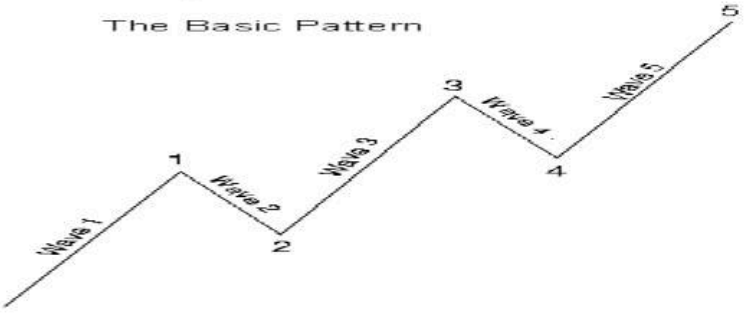
- Impulsive patterns
- Corrective patters

Let us discuss each of these patterns.

(a) **Impulsive Patterns-(Basic Waves) -** In this pattern there will be 3 or 5 waves in a given direction (going upward or downward). These waves shall move in the direction of the basic movement. This movement can indicate bull phase or bear phase.

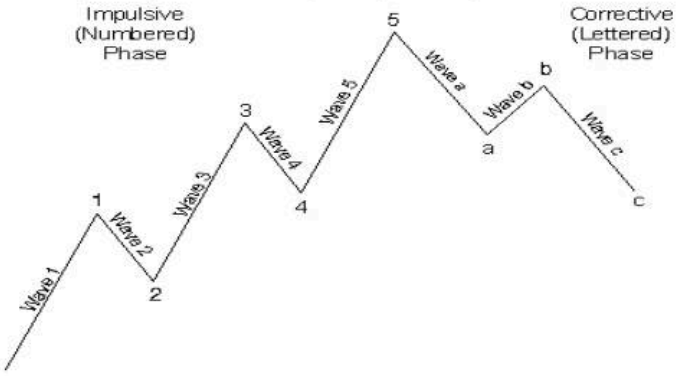
(b) **Corrective Patterns- (Reaction Waves) -** These 3 waves are against the basic direction of the basic movement. Correction involves correcting the earlier rise in case of bull market and fall in case of bear market.

As shown in the following diagram waves 1, 3 and 5 are directional movements, which are separated or corrected by wave 2 & 4, termed as corrective movements.



Source: http://elliotwave.net/

Complete Cycle -As shown in following figure five-wave impulses is following by a three-wave correction (a,b & c) to form a complete cycle of eight waves.



Source: http://elliotwave.net/

One complete cycle consists of waves made up of two distinct phases, bullish and bearish. On completion of full one cycle i.e. termination of 8 waves movement, the fresh cycle starts with similar impulses arising out of market trading.

ICAI answer: "Please refer paragraph 2.4.2" – SM paragraph 2.4.2 reproduced

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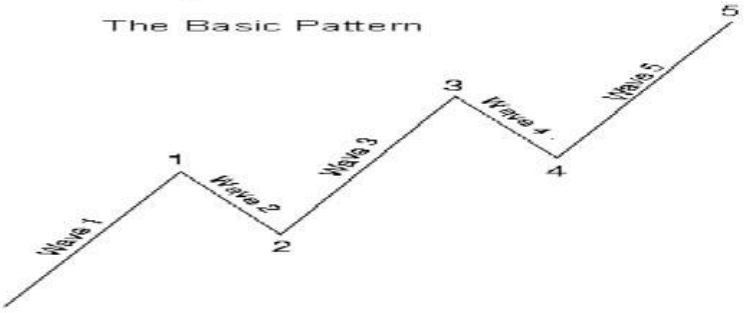
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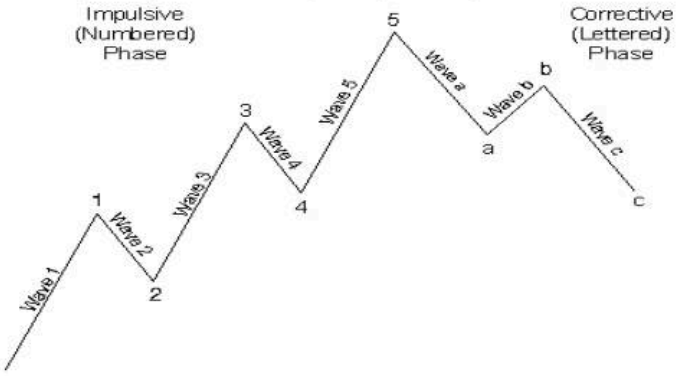
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Theory Q4

4. Explain the various indicators that can be used to assess the performance of an economy.

ICAI answer: "Please refer paragraph 1.2.2" – SM paragraph 1.2.2 reproduced

1.2.2 Techniques Used in Economic Analysis

Economic analysis is used to forecast national income with its various components that have a bearing on the concerned industry and the company in particular. Gross National Product (GNP) is used to measure national income as it reflects the growth rate in economic activities and has been regarded as a forecasting tool for analyzing the overall economy along with its various components during a particular period.

Some of the techniques used for economic analysis are:

(a) Anticipatory Surveys: They help investors to form an opinion about the future state of the economy. It incorporates expert opinion on construction activities, expenditure on plant and machinery, levels of inventory – all having a definite bearing on economic activities. Also future spending habits of consumers are taken into account.

In spite of valuable inputs available through this method, it has certain drawbacks:

(i) Survey results do not guarantee that intentions surveyed would materialize.

(ii) They are not regarded as forecasts per se, as there can be a consensus approach by the investor for exercising his opinion.

Continuous monitoring of this practice is called for to make this technique popular.

(b) Barometer/Indicator Approach: Various indicators are used to find out how the economy shall perform in the future. The indicators have been classified as under:

(i) *Leading Indicators*: They lead the economic activity in terms of their outcome. They relate to the time series data of the variables that reach high/low points in advance of economic activity.

(ii) *Roughly Coincidental Indicators:* They reach their peaks and troughs at approximately the same in the economy.

(iii) *Lagging Indicators:* They are time series data of variables that lag behind in their consequences vis-a- vis the economy. They reach their turning points after the economy has reached its own already.

All these approaches suggest direction of change in the aggregate economic activity but nothing about its magnitude. The various measures obtained form such indicators may give conflicting

signals about the future direction of the economy. To avoid this limitation, use of diffusion/composite index is suggested whereby combining several indicators into one index to measure the strength/weaknesses in the movement of a particular set of indicators. Computation of diffusion indices is no doubt difficult notwithstanding the fact it does not eliminate irregular movements.

Money supply in the economy also affects investment decisions. Rate of change in money supply in the economy affects GNP, corporate profits, interest rates and stock prices. Increase in money supply fuels inflation. As investment in stocks is considered as a hedge against inflation, stock prices go up during inflationary period.

(c) Economic Model Building Approach*:* In this approach, a precise and clear relationship between dependent and independent variables is determined. GNP model building or sectoral analysis is used in practice through the use of national accounting framework. The steps used are as follows:

(i) Hypothesize total economic demand by measuring total income (GNP) based on political stability, rate of inflation, changes in economic levels.

(ii) Forecasting the GNP by estimating levels of various components viz. consumption expenditure, gross private domestic investment, government purchases of goods/services, net exports.

(iii) After forecasting individual components of GNP, add them up to obtain the forecasted GNP.

(iv) Comparison is made of total GNP thus arrived at with that from an independent agency for the forecast of GNP and then the overall forecast is tested for consistency. This is carried out for ensuring that both the total forecast and the component wise forecast fit together in a reasonable manner.

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Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical QI

1. Closing values of NSE Nifty from 6th to 17th day of the month of January of the year 2020 were as follows:

Days	Date	Day	Sensex
1	6	THU	14522
2	7	FRI	14925
3	8	SAT	No Trading
4	9	SUN	No Trading
5	10	MON	15222
6	11	TUE	16000
7	12	WED	16400
8	13	THU	17000
9	14	FRI	No Trading
10	15	SAT	No Trading
11	16	SUN	No Trading
12	17	MON	18000

Calculate Exponential Moving Average (EMA) of Sensex during the above period. The previous day exponential moving average of Sensex can be assumed as 15,000. The value of exponent for 31 days EMA is 0.062.

Give detailed analysis on the basis of your calculations.

Solution — ICAI

1.

Date	1 Sensex	2 EMA for Previous day	3 1-2	4 3×0.062	5 EMA 2 ± 4
6	14522	15000	(478)	(29.636)	14970.364
7	14925	14970.364	(45.364)	(2.812)	14967.55
10	15222	14967.55	254.45	15.776	14983.32
11	16000	14983.32	1016.68	63.034	15046.354
12	16400	15046.354	1353.646	83.926	15130.28
13	17000	15130.28	1869.72	115.922	15246.202
17	18000	15246.202	2753.798	170.735	15416.937

Conclusion – The market is bullish. The market is likely to remain bullish for short term to medium term if other factors remain the same. On the basis of this indicator (EMA) the investors/brokers can take long position.

Practical Q2

2. The closing value of a Stock Market Index for the month of October, 2007 is given below:

Date Closing	Index Value
1.10.07	2800
3.10.07	2780
4.10.07	2795
5.10.07	2830
8.10.07	2760
9.10.07	2790
10.10.07	2880
11.10.07	2960
12.10.07	2990
15.10.07	3200
16.10.07	3300
17.10.07	3450
19.10.07	3360
22.10.07	3290
23.10.07	3360
24.10.07	3340
25.10.07	3290
29.10.07	3240
30.10.07	3140
31.10.07	3260

You are required to test the weak form of efficient market hypothesis by applying the run test at 5% and 10% level of significance.

Following values can be used:

Value of t at 5% is 2.101 at 18 degrees of freedom

Value of t at 10% is 1.734 at 18 degrees of freedom

Solution — ICAI

2.

Date	Closing Index	Sign of Price Charge
1.10.07	2800	
3.10.07	2780	-
4.10.07	2795	+
5.10.07	2830	+
8.10.07	2760	-
9.10.07	2790	+
10.10.07	2880	+
11.10.07	2960	+
12.10.07	2990	+
15.10.07	3200	+
16.10.07	3300	+
17.10.07	3450	+
19.10.07	3360	-
22.10.07	3290	-
23.10.07	3360	+
24.10.07	3340	-
25.10.07	3290	-
29.10.07	3240	-
30.10.07	3140	-
31.10.07	3260	+

Total of sign of price changes (r) = 8

No of Positive changes = $n_1 = 11$

No. of Negative changes = $n_2 = 8$

$$\mu_r = \frac{2n_1n_2}{n_1+n_2}+1$$

$$\mu = \frac{2\times11\times8}{11+8}+1 = 176/19 + 1 = 10.26$$

$$\hat{\sigma}_r = \sqrt{\frac{2n_1n_2(2n_1n_2-n_1-n_2)}{(n_1+n_2)^2(n_1+n_2-1)}}$$

$$\hat{\sigma}_r = \sqrt{\frac{(2\times11\times8)\,(2\times11\times8-11-8)}{(11+8)^2(11+8-1)}} = \sqrt{\frac{176\times157}{(19)^2(18)}} = \sqrt{4.252} = 2.06$$

Since too few runs in the case would indicate that the movement of prices is not random. We employ a two- tailed test the randomness of prices.

Test at 5% level of significance at 18 degrees of freedom using t- table

The lower limit

$= \mu - t \times \hat{\sigma}_r = 10.26 - 2.101 \times 2.06 = 5.932$

Upper limit

$= \mu + t \times \hat{\sigma}_r = 10.26 + 2.101 \times 2.06 = 14.588$

At 10% level of significance at 18 degrees of freedom

Lower limit

$= 10.26 - 1.734 \times 2.06 = 6.688$

Upper limit

$= 10.26 + 1.734 \times 2.06 = 13.832$

As seen r lies between these limits. Hence, the market exhibits weak form of efficiency.

*For a sample of size n, the t distribution will have n-1 degrees of freedom.

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